A complete C222 Diophantine equation

446 natural unknowns; degree 2,097,684. Unrendered layout plan.

Unknowns and front. Every free coordinate is in $\mathbb{N} = \{0, 1, 2, \ldots\}$:

$$h, e, g, \beta, F, x, N, \ell, z_0, z_t, z_B, \quad (\xi_i)_{i=0}^{8},$$
$$(D_i)_{i=0}^{3}, \quad (P_j)_{j=0}^{10}, \quad (r_j, s_j, w_j)_{j=0}^{10}, \quad (\theta_j)_{j=0}^{17}.$$

Each θ_j is the 21-tuple below. Every other name is a polynomial abbreviation; all bound indices are local to their displayed fixed ranges.

$$t = 16 + h,$$
$$(C, a, b, c, d, S, T, u, v) = (1 + \xi_i)_{i=0}^{8},$$
$$H = 2 + C + a + b + c + d + S + T + u + v.$$

The whole numerical calculus. Define

$$I(a,b) = 2(a+b)(a+b+1) + 4b,$$
$$M(a,b) = I(a,b) + 1, \qquad A(a,b) = I(a,b) + 2,$$
$$E(a,b) = I(a,b) + 3, \qquad \nu(a) = I(a,1),$$
$$X(a,b) = \nu(A(a,\nu(b))), \qquad \Theta_a(z) = \nu(A(a,z)).$$

The first eight code polynomials are

$$c_1 = X(a, E(a,b)),$$
$$c_2 = I(A(a, I(b,c)), I(A(a,b), A(a,c))),$$
$$c_3 = I(E(a,b), I(M(c,a), M(c,b))),$$
$$c_4 = I(A(a, A(b,c)), A(b, A(a,c))),$$
$$c_5 = I(E(c,a), I(M(c,b), M(a,b))),$$
$$c_6 = I(X(a, A(a,b)), b),$$
$$c_7 = I(E(c,a), I(E(c,b), A(M(a,b), E(a,b)))),$$
$$c_8 = I(I(I(c,d),a), I(I(a,c), I(b,c))).$$

The entire ninth code is

$$R(b,k,z) = A(b, I(z, \Theta_b(I(z, \nu(M(b,k)))))),$$
$$Q(a,b,k) = A(k, I(M(k,a), M(k,b))),$$
$$C_0(a,b,k,z) = I(Q(a,b,k), R(b,k,z)),$$
$$C_*(a,b,k,z) = A\big(b, I(M(b,k), \Theta_a(I(z, \Theta_a(C_0(a,b,k,z)))))\big),$$
$$V(a,b,c) = I(M(a,b), A(c, M(a,b))),$$
$$K(a,b,c) = A(c, I(C_*(a,b,c,c), \nu(R(b,c,c)))),$$
$$c_9 = \nu(I(V(a,b,c), K(a,b,c))).$$

Evaluate every c_i at (a,b,c,d), ignoring unused arguments. Put

$$W_z = 1 + Fz,$$
$$J = \sum_{i=1}^{9} c_i + a + b + I(a,b) + A(a,b) + 1,$$
$$G = ((c-a)^2(c-b)^2 - (d+1)^2)^2,$$
$$\mathcal{H} = (T-S)^2 \prod_{i=1}^{8} (TW_{c_i} - S)^2 \cdot \big((TW_{c_9} - S)^2 + G\big) \cdot (TW_b - SW_a W_{I(a,b)})^2 \cdot (TW_{A(a,b)} - SW_b)^2,$$
$$L_i = \mathcal{H} + (x - S - (1 + (i+1)y)u)^2 + (x - T - (1 + (i+2)y)v)^2.$$

Only L_C is used. The guarded branch retains both disequalities $c \neq a, c \neq b$ through its natural witness d and the printed G.

The 21-coordinate power polynomial. Every letter here is local to its copy, independent of the front and all other copies:

$$\theta = (Y, Z, X, U, S, R, a_0, y_0, f, d, b_+, b_-, s_+, s_-, r_+, r_-, x_+, x_-, g, w_n, w_k).$$

For arguments (n, k, m) define only

$$y = F + 1 + g, \quad A = a_0 + 2, \quad B = 1 + \beta, \quad V = Y^2(d+1),$$
$$B = 1 + 4Y(f+1), \qquad \tau = m + g + 1, \quad W = n + w_n,$$
$$\mathcal{N}(a,x,y) = x^2 + y^2 - 1 - a^2y^2.$$

In particular Y is free. Let $(R_1, \ldots, R_{11})$ be exactly

$$\big(Y - k - y_0,$$
$$\mathcal{N}(A,X,Y),\ \mathcal{N}(A,U,V),\ \mathcal{N}(B,S,R),$$
$$B + Ub_+ - A - Ub_-,\ S + Us_+ - X - Us_-,$$
$$R + 4Yr_+ - k - 4Yr_-,$$
$$X + Yn + \tau x_+ - YA - m - \tau x_-,$$
$$2An - \tau - n^2 - 1,\ W - k - w_k,\ \mathcal{N}(W+1, A, WZ)\big).$$

Set $\mathcal{P}(n,k,m;\theta) = \sum_{i=1}^{11} R_i^2$. All private coordinates, including both signs of every quotient, are natural.

Every argument slot. The full ordered table is

$$(k_i)_{i=0}^{3} = (t, t+1, y+1, x),$$
$$(U_j)_{j=0}^{10} = (F^2 + t, C, a, b, c, d, S, T, u, v, J),$$
$$(\sigma_j)_{j=0}^{10} = (0,0,1,1,1,1,2,2,3,3,1),$$
$$\mathcal{T} = \big((2(t+1), t, F), (B, x, N), (BH, x, q), (q, k_i, D_i)_{i=0}^{3},\ (q+1, U_j, P_j)_{j=0}^{10}\big).$$

For $0 \le j \le 17$, write $(n_j, k_j', m_j) = \mathcal{T}_j$ in this order and pair it with θ_j. No variable exponent occurs.

The single equation.

$$\begin{aligned}\mathcal{Q} = \sum_{j=0}^{17} \mathcal{P}(n_j, k_j', m_j; \theta_j)\\ + (x - F - 1 - (y+1)z_0)^2\\ + (x - 1 - (1 + (t+1)y)z_t)^2\\ + (Bz_B - C - F^2 - 1)^2 + (L_C - N\ell)^2\\ + (P_0 - D_0(qw_0 + B) - r_0)^2\\ + \sum_{j=1}^{10} (P_j - D_{\sigma_j} N w_j - r_j)^2\\ + \sum_{j=0}^{10} (D_{\sigma_j} - r_j - s_j - 1)^2 = 0.\end{aligned}$$

This prints the entire integer-coefficient polynomial: 224 squared residuals on 446 natural unknowns, of exact degree 2,097,684. For 446 positive unknowns, replace every free z independently by $p_z - 1$, $p_z \ge 1$. Relative to the retained ordinary Pell and C222/ZF/ZFC calibrations, absence of zeros is equivalent to syntactic Con(ZFC). No page fit has been tested.